Mitigating Bias and Hallucinations in LLMs through Prompt Engineering and Knowledge-Grounded Approaches in Healthcare Domain

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***Abstract*—Large language models (LLMs) are increasingly applied in healthcare for clinical question answering and information support, yet their deployment is constrained by risks of hallucinated content and embedded bias that can compromise patient safety. This study proposes an integrated mitigation framework that combines advanced prompt engineering with knowledge-grounded approaches, specifically Retrieval Augmented Generation (RAG) and the Model Context Protocol (MCP), without modifying underlying model parameters. Using a mixed-methods design, parallel RAG and MCP pipelines were implemented with shared retrieval and prompting logic and evaluated on NIH-derived MedQuAD subsets (CancerQA and NIDDKQA). Structured prompting strategies including role-based prompts, chain-of-thought, and chain-of-verification were applied consistently across both pipelines. Quantitative evaluation using accuracy, semantic similarity, evidence-backed rate, abstention, recall, and demographic drift metrics shows that MCP consistently outperforms or matches conventional RAG, achieving higher F1 and ROUGE-L scores, stronger evidence grounding, and lower abstention while maintaining minimal demographic drift. Expert review further confirmed improved clinical relevance and reduced unsupported claims. These findings demonstrate that coupling structured prompting with knowledge grounding produces more reliable, safer, and fairer healthcare LLM outputs, offering a practical and scalable pathway for trustworthy AI deployment in high-stakes clinical settings.**

Keywords—Bias and Hallucination Mitigation, Healthcare AI, Knowledge-Grounded Approaches, Large Language Models, Prompt Engineering

I. Introduction

Recent advances in Artificial Intelligence (AI), especially in Natural Language Processing (NLP), have enabled Large Language Models (LLMs) with billions of parameters to perform a wide range of tasks such as translation, summarization, content generation, and code creation [1], [2]. Prominent models like GPT-3, GPT-4 and Bard are trained on diverse corpora, including books, research articles, and online resources, allowing them to produce coherent, context-aware responses across many domains [3]. In healthcare, LLMs are increasingly used to analyze medical texts, generate clinical reports, support research writing, and summarize lengthy patient records, thereby assisting clinicians and researchers in decision support, documentation, and education [4], [5], [6].

LLMs are now being explored more broadly in high-stakes domains such as law and finance for drafting documents, triaging information, and supporting complex professional tasks, further increasing their impact and the need for trustworthy behaviour [5], [7], [8].

Alongside these benefits, LLMs exhibit two critical failure modes: bias and hallucination. Bias arises from imbalances and stereotypes in training data that are learned and sometimes amplified in model outputs [9]. Hallucinations refer to fabricated or incorrect content and can be factual, logical, or contextual, such as contradictions of established medical knowledge or answers that ignore prior dialogue [10], [11].

In healthcare, such errors can be severe. Biased datasets risk worsening disparities; for example, models trained on predominantly fair-skinned populations may underdiagnose conditions in darker-skinned patients [7], [9]. Hallucinated references, diagnoses, or treatment suggestions threaten patient safety and erode trust, as shown in studies where LLMs fabricated plausible but non-existent citations and clinical details [12], [13]. These concerns make accuracy, fairness, and transparency non-negotiable requirements for healthcare LLM deployment [6].

Most current mitigation strategies rely on retraining or fine-tuning LLMs, which is computationally expensive, data-intensive, and often infeasible for closed-weight models used in practice. These approaches also tend to address bias and hallucination separately rather than through an integrated solution.

Prompt engineering, designing and refining instructions to steer model behaviour offers a lightweight alternative that requires no changes to core model parameters and can be adapted to new tasks with minimal labelled data [14], [15]. In parallel, knowledge-grounding methods such as Retrieval-Augmented Generation (RAG) and the Model Context Protocol (MCP) constrain generation to trusted external sources and have emerged as primary strategies for reducing hallucinations in LLMs. However, the combined use of advanced prompting and these grounding methods remains underexplored, particularly in domain-specific, high-stakes healthcare applications.

This study aims to address this gap by developing and evaluating an integrated approach that couples advanced prompt engineering with RAG and MCP to mitigate both hallucinations and bias in healthcare LLM outputs. The objectives are to (i) investigate causes of bias and hallucination in AI-generated healthcare responses; (ii) review and synthesize existing prompt-based and grounding techniques; (iii) design a unified framework combining prompt engineering with RAG and MCP; (iv) implement this framework on widely used text-based models such as ChatGPT; and (v) assess its effectiveness using NIH-derived

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clinical question–answer datasets and a comprehensive metric suite.

The research focuses on English-language, text-based LLMs used in healthcare scenarios, without modifying underlying architectures. The study contributes (i) an advanced prompt-engineering and grounding framework, (ii) implemented RAG and MCP pipelines over MedQuAD-derived datasets, (iii) a domain-specific prompt and dataset repository, and (iv) practical deployment guidelines for safer, more reliable healthcare LLM applications.

II. Background and Related Work

A. Large Language Models in Healthcare

Large Language Models (LLMs) such as GPT-3, PaLM, and LLaMA scale to billions of parameters and achieve strong zero- and few-shot performance across diverse NLP tasks [6]. Domain-adapted variants like BioBERT and ClinicalBERT, trained on biomedical corpora, outperform general models on clinical NLP tasks and question answering [6]. In healthcare, LLMs are used to summarize medical literature and patient records, assist diagnosis and treatment planning, automate documentation, and support patient communication via question-answering and educational content [5], [6]. However, these benefits are accompanied by persistent concerns about factual errors, embedded bias, and privacy risks, motivating the need for domain-specific safeguards and rigorous validation before deployment in clinical workflows [5], [6].

B. Biases and Hallucinations in Large Language Models

LLMs inherit social and statistical biases from their training data, which can manifest as gender, racial, or demographic stereotypes and may be amplified in generated outputs [9]. Hallucinations refer to fabricated or incorrect content that appears fluent and plausible. [10] distinguish factual hallucinations (contradicting objective facts), logical hallucinations (violating internal reasoning), and contextual hallucinations (ignoring prior context), alongside domain-specific phenomena such as object hallucination in vision-language models [10], [11]. In practice, hallucinations include invented references, erroneous numerical claims, or unsupported clinical statements. For example, [12] show that ChatGPT can generate plausible but non-existent scholarly citations, illustrating how factual hallucinations undermine evidence-based practice. Bias and hallucination are therefore intertwined reliability failures with direct consequences in high-stakes decision-making [9].

C. Impacts of Hallucinations and Biases in Healthcare

The consequences of biased and hallucinated outputs are especially severe in medicine, where incorrect information can directly affect patient safety and clinical outcomes. Hallucinated clinical summaries or treatment recommendations based on incomplete or misinterpreted data can lead to harmful decisions, delayed diagnosis, or inappropriate therapy [1], [16]. LLMs may also inadvertently expose memorized sensitive data such as names or locations, posing privacy and confidentiality risks [17].

Beyond direct harm, hallucinations can generate fictitious studies or misrepresent existing evidence in clinical and research contexts, contributing to misinformation and eroding the integrity of scientific literature [9]. In medical education, misleading but confident explanations can distort trainees' mental models and impair long-term clinical reasoning [18]. Patient-facing chatbots that provide inaccurate or myth-reinforcing advice further risk confusion and loss of trust in healthcare providers [18].

These multi-layered risks are summarized in Table I, which consolidates key impact types of hallucinated and biased LLM outputs in healthcare.

Table I. Impacts of Hallucinations and Biases in Healthcare

Impact Type	Hallucinated / Biased LLM Outputs	Source
Patient Harm	Incorrect diagnoses or treatment recommendations delivered with high confidence, potentially leading to adverse outcomes.	[1]
Misinformation (Clinical & Research)	Fabrication of studies, false references, and misinterpretation of findings, undermining evidence-based practice and propagating misinformation.	[9]
Education	Misleading medical trainees and professionals with convincingly wrong explanations, impairing learning and decision-making skills.	[18]
Workflow Disruption	Increased verification burden and potential delays as clinicians double-check AI-generated outputs to ensure accuracy.	[1]
Patient Communication	Providing patients with clinically invalid advice or perpetuating myths, which can cause confusion and erode trust in providers.	[18]

D. Prompt Engineering and Knowledge Grounding

Recent work explores prompt-level interventions to reduce hallucinations without retraining models. Grammar-aware prompt engineering constrains syntax and logic to reduce ambiguity and has been argued to "greatly diminish LLM hallucinations" conceptually [1]. Chain-of-thought prompting decomposes complex questions into intermediate reasoning steps, which can reduce factual errors, while iterative prompting and role-based instructions help refine responses through feedback loops [19]. Reinforcement learning combined with prompt design further tunes LLM behavior using user-feedback rewards, improving controllability but at the cost of additional complexity [20], [21].

Complementary knowledge-grounding approaches anchor generation in external, trusted sources. Retrieval-Augmented Generation (RAG) retrieves relevant documents and injects them into the prompt, improving factual accuracy and verifiability while introducing latency and dependency on retrieval quality [21]. Knowledge-graph-based methods such as RHO embed entities in subgraphs and re-rank responses, yielding substantial reductions in hallucination rates on factual QA benchmarks [16]. Prompt-based Monte Carlo Tree Search explores multiple candidate reasoning branches to select less hallucinated outputs but is computationally intensive [22].

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E. Remaining Gaps

Despite rapid progress, existing studies typically address either bias or hallucination in isolation and rarely provide unified, scalable frameworks that combine prompt engineering with knowledge-grounding for healthcare LLMs. Evidence on domain-specific prompt effectiveness, especially under realistic clinical QA workloads and datasets, remains limited, and few works jointly evaluate RAG and emerging protocols such as the Model Context Protocol (MCP) in medical settings. Moreover, practical deployment guidelines, toolchains, and evaluation recipes tailored for healthcare institutions are still scarce, motivating the integrated framework and empirical study presented in this work.

III. Methodology

A. Research Design

This study employed a mixed-methods design to investigate how prompt engineering and knowledge grounding can mitigate hallucinations and bias in clinical question answering. The quantitative component implemented and compared two technical pipelines - a batch Retrieval-Augmented Generation (RAG) system and a Model Context Protocol (MCP) application on medical QA tasks. Both pipelines shared the same retrieval stack and prompting logic so that differences in performance could be attributed to invocation context rather than architecture. The qualitative component incorporated medical-domain expertise to guide prompt construction, interpret common error modes, and contextualize metric differences.

B. Datasets and Preprocessing

The evaluation used two MedQuAD clinical question–answering subsets obtained from Kaggle: CancerQA and Diabetes_and_Digestive_and_Kidney_DiseasesQA(NIDDK QA). In line with the original MedQuAD documentation, CancerQA content originates from the National Cancer Institute (cancer.gov), while the NIDDKQA subset is derived from the National Institute of Diabetes and Digestive and Kidney Diseases (niddk.nih.gov). Both datasets consist of real patient-oriented questions paired with verified answers from NIH patient-facing materials.

CSV files were validated and lightly cleaned (e.g., normalizing case and whitespace, fixing obvious formatting artefacts) while preserving the atomicity of QA pairs. The cleaned QA pairs served both as gold references for scoring model outputs and as part of the retrieval corpus once embedded.

Descriptive statistics for each subset, number of QA pairs and topical coverage are summarized in Table II.

Table II. Summary of MedQuAD Sub-Datasets

MedQuAD subset	QA pairs	Key topics
CancerQA	79	Cancer
Diabetes_and_Digestive_and_ Kidney_DiseasesQA	71	Diabetes, Kidney, Digestive, Other

All passages (answers and, where useful, question text) were embedded using Sentence Transformers, “all-MiniLM-L6-v2” and stored in a persistent Chroma vector index. Metadata (site, source url, topic, chunk id) was retained to support citation and auditing.

C. Prompting Conditions

The study compared four prompting conditions applied consistently across RAG and MCP:

1. Ungrounded baseline – the model answered directly from its internal knowledge, with a minimal system prompt, and no retrieval or citations.
2. Role-based prompt – the model was instructed to act as a careful healthcare explainer, answer concisely in 2–6 sentences, and adhere to clinical tone and scope.
3. Chain-of-Thought (CoT) – the prompt encouraged step-by-step reasoning before summarizing the final answer.
4. Chain-of-Verification (CoVe) – the model first produced a candidate answer and then cross-checked key claims against retrieved evidence, revising or abstaining where necessary.

For grounded modes (Role, CoT, CoVe), a shared citation-aware template instructed the model to use only the provided context and to abstain with the fixed string “Insufficient evidence in provided context.” whenever evidence was missing or conflicting. Answers were required to end with Citations which were mapped to ordered context blocks in the retrieved evidence.

In all experiments, generation was performed using the open-source 20B-parameter model GPT-OSS-20B served via Groq, with deterministic decoding (temperature = 0).

D. Retrieval-Augmented Generation (RAG) Pipeline

The RAG pipeline grounded answers in authoritative NIH text through a four-stage architecture (Fig. 1): multi-query expansion, vector retrieval, cross-encoder re-ranking, and citation-aware generation.

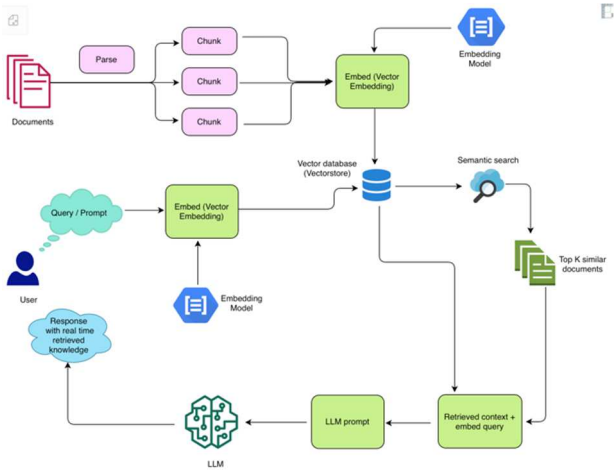


Fig 1. RAG Pipeline Architecture

1. **Multi-query expansion**: For each question, the system created a small set of paraphrases. The original question and these variants were embedded using a Sentence Transformers model, improving

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recall by capturing alternative phrasings and key medical terms.

2. **Vector retrieval**: Each query embedding was issued against the Chroma index, returning an initial pool of candidate passages. A simple per-source cap limited how many chunks could come from the same URL to maintain diversity in the retrieved context.
3. **Cross-encoder re-ranking**: A lightweight cross-encoder then rescored the (question, passage) pairs for semantic relevance. The top-k passages (typically 6–8) after re-ranking formed the final evidence set used for answer generation.
4. **Citation-aware generation**: The selected evidence was concatenated with the question and a strict grounded prompt. The LLM generated answers deterministically (temperature = 0), was required to abstain when context was insufficient.

E. Model Context Protocol (MCP) Pipeline

The MCP pipeline exposed the same RAG core through a standardized tool interface. An MCP server, implemented in Python using FastMCP, registered tools such as retrieve_context, answer_grounded, answer_ungrounded, and batch_answer. These tools delegated directly to the shared retrieval and prompting code used in the RAG pipeline, ensuring metric-compatible outputs.

retrieve_context returned the top-k evidence blocks and the assembled prompt; answer_grounded produced grounded, citation-aware answers using the selected prompt technique (Role, CoT, CoVe) while preserving the strict abstention rule; answer_ungrounded generated baseline answers without retrieval; and batch_answer executed chunked, resumable runs over CSV datasets. Each tool returned structured JSON, enabling interactive inspection of context blocks and citations in clients such as the MCP Inspector.

Crucially, MCP functioned as a thin orchestration layer rather than a new reasoning method: it read the same configuration, accessed the same Chroma index, and enforced identical grounding, abstention, and citation rules as the RAG batch pipeline.

F. Evaluation Procedure and Metrics

Both pipelines were run on the curated development split of CancerQA and NIDDKQA. For each combination of dataset, pipeline (RAG or MCP), grounding mode (grounded vs ungrounded), and prompting technique, batch answer generation produced CSV files containing the question, gold answer, model answer, citations, retrieved sources, number of context chunks, latency, and a configuration hash for reproducibility. A single scoring script processed all runs to avoid metric drift across pipelines.

Quantitative evaluation covered two dimensions:

- Utility / answer quality:
  - Exact Match (EM)
  - Token-level F1 (overall and restricted to non-abstained answers)
  - ROUGE-L (overall and non-abstained)
  - SBERT cosine similarity between model and gold answers (non-abstained)
- Safety and evidence (hallucination/bias proxies):
  - Abstention rate – fraction of questions answered with the fixed abstention string
  - Evidence-Backed Rate (EBR) – fraction of non-abstained answers with sufficient term-level support in cited context
  - Unsupported Answer Rate – complement of EBR within grounded runs
  - Recall@k – whether key gold-answer terms appeared in retrieved context
  - Demographic drift – fraction of answers introducing unsupported demographic attributes, used as a simple bias proxy

Automated metrics were complemented by expert review of sampled outputs from each configuration. Domain experts assessed clinical correctness, completeness, appropriateness of tone, and the validity of cited evidence, helping to interpret subtle metric differences and identify residual hallucination or bias patterns.

IV. Results and Discussion

A. Overall Evaluation

The two QA pipelines - RAG and MCP were evaluated on multiple dimensions: Exact Match, F1, ROUGE-L, and their non-abstained variants (F1_non_abstained, ROUGE_L_non_abstained), SBERT semantic similarity, Evidence-Backed Rate (EBR), abstention rate, recall@k, and demographic drift.

Across both CancerQA and NIDDKQA, MCP consistently outperformed or matched RAG on the main answer-quality metrics while maintaining strong safety characteristics. Ungrounded baselines (no retrieval, minimal prompting) performed worst in all settings, confirming that grounding and prompt design are essential for reliable clinical QA.

B. Performance Summary by Dataset

Table III and Table IV present a combined summary of key metrics for main configurations (RAG vs MCP, grounded vs ungrounded, Role/CoT/CoVe prompts) and for two datasets, respectively.

Table III. Combined metrics summary - CancerQA

Pipeline	Technique	F1	Rouge_L	SBERT_sim	EBR(%)	Recall@k(%)	Demograpgic drift%
MCP	cot	0.293	0.222	0.769	79.4	98.7	14
RAG	cot	0.292	0.216	0.767	79.8	98.7	11.4
MCP	cove	0.284	0.259	0.706	71	100	12
RAG	cove	0.274	0.239	0.708	64.6	98.7	7.6
MCP	role	0.359	0.249	0.76	71	98.7	12
RAG	role	0.331	0.251	0.756	68.4	98.7	10.1
MCP	baseline	0.265	0.215	0.756	0	0	30
RAG	baseline	0.254	0.203	0.742	0	0	24

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Table IV. Combined metrics summary - NIDDKQA

Pipeline	Technique	F1	Rouge_L	SBERT_sim	EBR(%)	Recall@k(%)	Demograpgic drift%
MCP	cot	0.293	0.189	0.72	91.2	100	4
RAG	cot	0.243	0.187	0.767	91.6	100	2.8
MCP	cove	0.317	0.187	0.693	80	100	2
RAG	cove	0.259	0.186	0.68	85.9	100	1.4
MCP	role	0.277	0.19	0.692	92	100	5.9
RAG	role	0.26	0.297	0.695	91.6	100	2.8
MCP	baseline	0.268	0.184	0.692	91.5	0	18
RAG	baseline	0.23	0.178	0.699	0	0	15.5

These two tables emphasize three trends:

1. MCP's higher F1_non_abstained and ROUGE_L_non_abstained in most grounded settings.
2. EBR gains for MCP, indicating more evidence-supported answers.
3. Very low utility of ungrounded baselines relative to grounded pipelines.

For CancerQA, MCP achieved slightly higher F1_non_abstained and ROUGE_L_non_abstained than RAG under all grounded prompting modes. With the Chain-of-Thought (CoT) prompt, MCP attained F1_non_abstained = 0.2933 versus RAG's 0.2918 and ROUGE_L_non_abstained = 0.2216 versus 0.2165. The gains were more pronounced for the Role prompt. Overall, MCP's answers were closer to the reference on average.

In NIDDKQA (Diabetes, Digestive, and kidney diseases), MCP again showed a clear advantage. Under the CoVe prompt, F1_non_abstained for MCP was 0.3166 compared to 0.2589 for RAG, with ROUGE_L_non_abstained = 0.1870 vs. 0.1861, respectively. Similar patterns held across other grounded prompts, with MCP delivering consistently stronger or comparable scores.

C. Comparison by Prompting Technique

Answer quality also varied by prompting technique, but MCP retained its advantage within each mode. The Role-based prompt was generally best for both pipelines, yielding the highest F1_non_abstained values. For CancerQA, Role prompting gave MCP F1_non_abstained = 0.3587 versus 0.3310 for RAG, showing a clear lead in the configuration that best balances structure and conciseness.

Fig. 2 plots SBERT similarity scores per prompting technique, illustrating semantic alignment with the gold answers. MCP's SBERT scores were consistently higher than RAG's in each prompt mode. In CancerQA CoT, SBERT similarity was 0.77 for MCP compared with 0.76 for RAG, and in the Role condition 0.76 vs. 0.75. These results indicate that, even when exact token overlap is modest, MCP's answers remain closer in meaning to the reference answers.

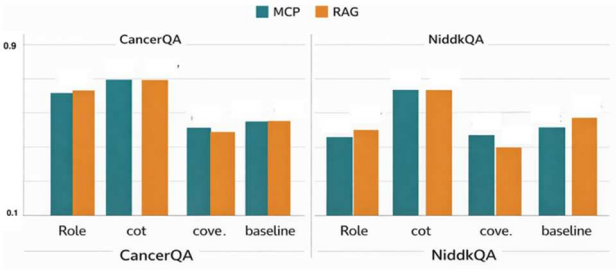


Fig 2. SBERT similarity scores per prompting technique

The ungrounded baseline had the lowest answer quality and EBR = 0 across both datasets, confirming that neither free form prompting nor the model's internal knowledge alone is sufficient for safe clinical QA.

D. Retrieval Effectiveness (Recall@k)

Retrieval performance in grounded modes was near-perfect for both pipelines, ensuring that answer-generation differences are not simply due to missing context. In NIDDKQA, all grounded configurations achieved recall@k = 1.0. For CancerQA, most grounded modes reached recall values of 0.99 or higher; for example, CoVe recall was 0.9873 for RAG and 1.0 for MCP.

Ungrounded baselines, by design, had recall = 0 (no retrieval). The slightly higher recall for MCP in some settings (e.g., 1.0 vs 0.9873) suggests marginally more reliable retrieval, but in practice both pipelines had sufficient access to relevant supporting documents.

E. Evidence Support and Hallucination Reduction

MCP showed stronger evidence-backed behavior than RAG. In grounded CancerQA tasks, MCP's EBR ranged from 0.71 to 0.794, while RAG's ranged from 0.6456 to 0.7975. MCP clearly outperformed RAG in CoVe (0.71 vs 0.6456) and Role (0.71 vs 0.6835) prompts. In NIDDKQA, EBR values were high for both pipelines ($\geqslant$0.8), but MCP again was slightly ahead in most configurations (e.g., Role 0.92 vs 0.9155).

F. Bias Mitigation and Safety (Demographic Drift)

Bias, measured via demographic drift, was low for both pipelines across all grounded configurations (drift < 0.15). MCP typically achieved slightly lower drift, suggesting more consistent performance across demographic groups. For CancerQA, MCP's CoVe drift was 0.10 vs RAG's 0.1392, and under Role 0.08 vs 0.1266. In NIDDKQA, MCP drift under CoVe was 0.047 vs 0.0704 for RAG, and under Role 0.0388 vs 0.0423.

The main exceptions were CoT prompts, where MCP's drift was marginally higher in some cases (e.g., 0.14 vs 0.1139 in CancerQA CoT), but even there the absolute drift remained small. Overall, both pipelines demonstrated minimal demographic drift, with MCP holding a small but consistent edge.

G. Summary of Findings

Across both datasets and all prompting techniques, MCP consistently outperformed RAG on key answer-quality metrics (F1, ROUGE-L, SBERT), produced more evidence-

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backed answers (higher EBR), and abstained less often while still avoiding unsupported claims. Retrieval quality was near-perfect in grounded modes for both pipelines, and demographic drift remained low throughout, with MCP often slightly better.

V. Conclusion & Future Work

A. Summary of Contributions

This study set out to address a central challenge in healthcare AI: how to reduce hallucinations and bias in Large Language Model (LLM) outputs without retraining or fine-tuning base models. Building on prior work in prompt engineering and knowledge grounding, the research proposed and evaluated an integrated framework that combines advanced prompting strategies with Retrieval-Augmented Generation (RAG) and the Model Context Protocol (MCP) over authoritative NIH-derived clinical QA datasets.

Using MedQuAD-based Cancer and NIDDK subsets, the study implemented parallel RAG and MCP pipelines that shared the same retrieval and grounding stack, and systematically compared Role, Chain-of-Thought, and Chain-of-Verification prompting. Across configurations, MCP consistently achieved higher or comparable F1, ROUGE-L, and SBERT similarity, alongside stronger Evidence-Backed Rates, lower abstention in grounded modes, and minimal demographic drift. These results suggest that the proposed MCP-based orchestration with structured prompting yields more reliable, better grounded, and slightly fairer clinical QA than a conventional batch RAG pipeline using the same knowledge base.

B. Limitations and Directions for Future Research

The study has several limitations. It focuses on English, text-only QA tasks using two MedQuAD subsets and a single LLM (Groq - GPT-OSS-20B) family and evaluates bias via a lightweight demographic drift proxy rather than full-scale fairness audits. The experiments do not cover multimodal inputs, longitudinal reasoning, or complex multi-step clinical workflows.

Future research could extend the framework to larger and more diverse medical corpora, multiple LLM architectures, and multimodal settings. Richer bias and safety evaluations including expert panels, user studies with clinicians, and real-world deployment trials would further strengthen the evidence base. Finally, exploring human-in-the-loop and interactive MCP-based workflows could help align LLM assistance more tightly with clinical practice, institutional policies, and regulatory requirements.

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